One unit, two integer coordinates

n	x(n)	y(n)	norm
0	1	0	1
1	2	1	1
2	7	4	1
3	26	15	1
4	97	56	1
5	362	209	1
6	1351	780	1
7	5042	2911	1

$$x(n+1) = 2x(n) + 3y(n)$$
$$y(n+1) = x(n) + 2y(n)$$

$(7, 4) \rightarrow (26, 15)$

determinant = 1

norm $x^2 - 3y^2 = 1$